MTH 211: Theory of Statistics.
2025-26 (II)

Lecture 17: Maximum Likelihood Estimator (MLE) – More Examples

Ex. 3 $X_1, \dots, X_n$ r.s. from $\mathcal{E}xp(1/\theta)$, $\theta > 0$

$$L(\theta|\underset{\sim}{x}) = \theta^n e^{-\theta \Sigma x_i}, \quad l(\theta|\underset{\sim}{x}) = n \ln\theta - \theta \Sigma x_i$$

$$\left.\frac{\partial l(\theta|\underset{\sim}{x})}{\partial \theta}\right|_{\hat\theta} = \frac{n}{\hat\theta} - \Sigma x_i = 0 \Rightarrow \hat\theta = \frac{n}{\Sigma x_i} = \frac{1}{\bar x}$$

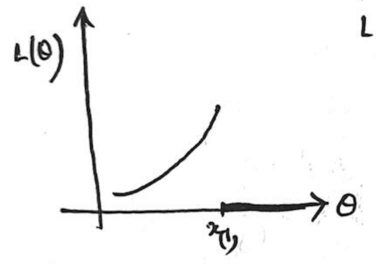
$$\left.\frac{\partial^2 l(\theta|\underset{\sim}{x})}{\partial \theta^2}\right|_{\hat\theta} = -\frac{n}{\hat\theta^2} < 0 \qquad \therefore \hat\theta = \frac{1}{\bar X} \text{ is MLE of } \theta.$$

Ex. 4 $X_1, \dots, X_n$ r.s. from $\mathcal{E}xp(\theta, 1)$, $\theta \in \mathbb{R}$

$$\cancel{L(\theta|\underset{\sim}{x}) = e^{-\Sigma(x_i-\theta)} = e^{n\theta} e^{-\Sigma x_i} I(\theta, x_{(1)}), \; x_i > \theta \; \forall i}$$

$$f_\theta(\underset{\sim}{x}) = e^{-\Sigma(x_i-\theta)}, \quad x_i > \theta \; \forall i$$

$$L(\theta|\underset{\sim}{x}) = e^{n\theta} e^{-\Sigma x_i} I(\theta, x_{(1)}), \quad \text{fn is not diff'ble } \forall \theta \in \mathbb{R}.$$

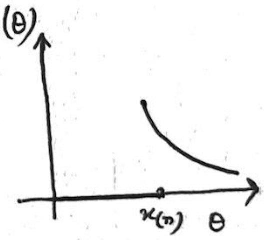


$$\begin{cases} = 0 & \text{if } \theta > x_{(1)} \\ > 0 \text{ and } \uparrow \theta & \text{if } \theta \le x_{(1)} \end{cases}$$

$$\Rightarrow \hat\theta = X_{(1)} \text{ is MLE of } \theta.$$

Ex. 5. $X_1, \dots, X_n$ r.s. from $U[0, \theta]$, $\theta > 0$

$$L(\theta|\underset{\sim}{x}) = \frac{1}{\theta^n} I(x_{(n)}, \theta) \begin{cases} = 0 & \text{if } \theta < x_{(n)} \\ > 0 \text{ \& } \downarrow \theta & \text{if } \ge \end{cases}$$



$$\hat\theta = X_{(n)} \text{ mle of } \theta.$$

Ex. 6. MLE not unique case.
$X_1, \dots, X_n$ r.s. from $U[\theta - 1/2, \theta + 1/2]$, $\theta \in \mathbb{R}$.

$$L(\theta|\underset{\sim}{x}) = 1 \cdot I_{(\theta - 1/2, x_{(1)})} I_{(x_{(n)}, \theta + 1/2)}$$

$$= \begin{cases} 1 & \text{if } x_{(n)} - 1/2 \le \theta \le x_{(1)} + 1/2 \\ 0 & \text{ow} \end{cases}$$

Hence any $\hat\theta \ni X_{(n)} - 1/2 \le \hat\theta \le X_{(1)} + 1/2$ is MLE of θ,
e.g. $\forall \alpha \in [0,1]$, $T_\alpha(\underset{\sim}{x}) = \alpha(X_{(1)} + 1/2) + (1-\alpha)(X_{(n)} - 1/2)$ is an MLE of θ.
If $\alpha = 1/2$, $T(\underset{\sim}{x}) = \frac{X_{(1)} + X_{(n)}}{2} = \hat\theta_{mle}$.

Ex 7. Multiparameter case.

$x_1, \dots, x_n$ r.s. from $N(\mu, \sigma^2)$, $\underset{\sim}{\theta} = (\mu, \sigma^2)' \in \Theta \subseteq \mathbb{R}^2$

$$L(\underset{\sim}{\theta}|\underset{\sim}{x}) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$l(\underset{\sim}{\theta}|\underset{\sim}{x}) = \text{const} - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_i (x_i-\mu)^2$$

$$\left.\frac{\partial l(\underset{\sim}{\theta}|\underset{\sim}{x})}{\partial\mu}\right|_{\hat\mu,\hat\sigma^2} = \cancel{-\frac{n}{2}\cdot\frac{1}{\hat\sigma^2}} \quad \frac{2\sum(x_i-\hat\mu)}{2\hat\sigma^2} = 0 \Rightarrow \hat\mu = \bar{x}$$

$$\left.\frac{\partial l(\underset{\sim}{\theta}|\underset{\sim}{x})}{\partial\sigma^2}\right|_{\hat\mu,\hat\sigma^2} = -\frac{n}{2}\cdot\frac{1}{\hat\sigma^2} + \frac{1}{2}\cdot\frac{1}{(\hat\sigma^2)^2}\sum(x_i-\hat\mu)^2 \Rightarrow \hat\sigma^2 = \frac{1}{n}\sum(x_i-\bar{x})^2$$

Calculations for the Hessian matrix:

$$\left.\frac{\partial^2 l(\underset{\sim}{\theta}|\underset{\sim}{x})}{\partial\mu^2}\right|_{\hat\mu,\hat\sigma^2} = -\frac{n}{\hat\sigma^2}, \quad \left.\frac{\partial^2 l(\theta|\underset{\sim}{x})}{\partial\mu\,\partial\sigma^2}\right|_{\hat\mu,\hat\sigma^2} = \left.\frac{\partial^2 l(\theta|\underset{\sim}{x})}{\partial\sigma^2\,\partial\mu}\right|_{\hat\mu,\hat\sigma^2} = 0$$

$$\left.\frac{\partial^2 l(\theta|\underset{\sim}{x})}{\partial(\sigma^2)^2}\right|_{\hat\mu,\hat\sigma^2} = \frac{n}{2}\cdot\frac{1}{(\hat\sigma^2)^2} - \frac{\sum(x_i-\hat\mu)^2}{(\hat\sigma^2)^3} = -\frac{n}{2(\hat\sigma^2)^2}.$$

so that $H = \begin{pmatrix} -n/\hat\sigma^2 & 0 \\ 0 & -n/2\hat\sigma^4 \end{pmatrix}$, which is nnd.

$\Rightarrow \hat\theta = \begin{pmatrix}\hat\mu \\ \hat\sigma^2\end{pmatrix} = \begin{pmatrix}\bar{X} \\ \frac{1}{n}\sum(X_i-\bar{X})^2\end{pmatrix}$ is mle of θ,

$\Rightarrow$ Separately (or, individually) $\hat\mu = \bar{X}$ and $\hat\sigma^2 = \frac{1}{n}\sum(X_i-\bar{X})^2$ are MLE's of μ and σ^2.

Ex 8. Multiparameter case w/o a closed form solution of the likelihood or score eqns.

$x_1, \dots, x_n$ r.s. from Gamma(α, β), $\alpha, \beta > 0$.

$$L(\underset{\sim}{\theta}|\underset{\sim}{x}) = \frac{1}{(\beta^\alpha\Gamma\alpha)^n}\prod_{i=1}^{n} x_i^{\alpha-1}\, e^{-1/\beta\sum_1^n x_i}$$

$$l(\underset{\sim}{\theta}|\underset{\sim}{x}) = -n\alpha\ln\beta - n\ln\Gamma\alpha + (\alpha-1)\sum_i \ln x_i - \sum x_i/\beta$$

$\left(\Gamma'\alpha = \frac{\partial}{\partial\alpha}\Gamma\alpha\right.$, the digamma fn$\left.\right)$

$$\frac{\partial l(\theta)}{\partial\alpha} = -n\ln\beta - n\frac{\Gamma\alpha'}{\Gamma\alpha} + \sum\ln x_i = 0$$

$$\frac{\partial l(\theta)}{\partial\beta} = -n\alpha/\beta + \frac{\sum x_i}{\beta^2} = 0, \quad \because \beta = \frac{\bar{x}}{\alpha}$$

, use this in the 1st eqn & solve numerically for $\hat\alpha$ and hence to get $\hat\alpha$ get $\hat\beta = \bar{x}/\hat\alpha$.

Check 2nd derivative condns numerically to confirm MLE.